

REPORT

TOPIC:- HOSPITAL MANAGEMENT SYSTEM

Problem statement:-

- Hospitals often face challenges in managing patient records, doctor schedules, and appointments efficiently. Manual systems lead to errors, delays, and poor coordination. A digital solution is required to streamline hospital operations, reduce human error, and improve patient care.

Objectives:-

- To design a Java-based system that manages patients, doctors, and appointments.
- To demonstrate Object-Oriented Programming concepts in a real-world application.

- To provide a user-friendly, menu-driven interface for hospital staff.
- To enable quick retrieval of patient and doctor information.

Scope:-

- Patient Management: Add and view patient details.
- Doctor Management: Add and view doctor details.
- Appointment Management: Book and view appointments.
- Console-based interface with user input.
- Future scope includes file handling, database integration, and GUI.

System workflow:-

- **Patient** → Stores patient details (ID, name, age, disease).
- **Doctor** → Stores doctor details (ID, name, specialization).
- **Appointment** → Links patient with doctor.
- **HospitalManagementSystem** → Main class with menu-driven interface.

Implementation:-

- Language: **Java SE**
- Input: **Scanner class** for user input.
- Storage: **ArrayList** for dynamic data handling.

- Concepts: Encapsulation, Abstraction, Polymorphism.

Sample output:-

```
🏥 Hospital Management System
1. Add Patient
2. Add Doctor
3. Book Appointment
4. View Patients
5. View Doctors
6. View Appointments
7. Exit
Enter your choice: 1
Enter Patient ID: 102
Enter Patient Name: saransh
Enter Age: 19
Enter Disease: fever
✅ Patient added successfully!
```

```
🏥 Hospital Management System
1. Add Patient
2. Add Doctor
3. Book Appointment
4. View Patients
5. View Doctors
6. View Appointments
7. Exit
Enter your choice: 1
Enter Patient ID: 102
Enter Patient Name: deepak
Enter Age: 18
Enter Disease: typhoid
✅ Patient added successfully!
```

```
🏥 Hospital Management System
1. Add Patient
2. Add Doctor
3. Book Appointment
4. View Patients
5. View Doctors
6. View Appointments
7. Exit
Enter your choice: 6

📅 Appointments List:
Appointment ID: 203, Patient: saransh, Doctor: rahul
```

Advantages:-

- Easy to use and extend.

- Demonstrates core Java concepts.
- Reduces manual errors in hospital management.
- Can be scaled to include GUI and database.

Future enhancements:-

- File handling for persistent storage.
- Database integration (MySQL + JDBC).
- GUI interface using Swing/JavaFX.
- Advanced features like billing, prescriptions, and reports.

Conclusion:-

The Hospital Management System provides a simple yet effective solution for managing hospital operations. It demonstrates how Java can be applied to real-world problems using OOP concepts. With future enhancements, it can evolve into a complete hospital ERP solution.